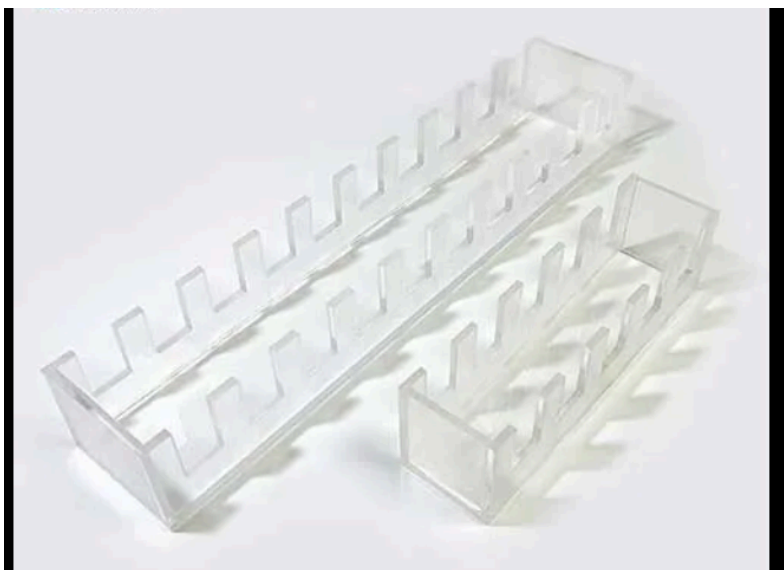
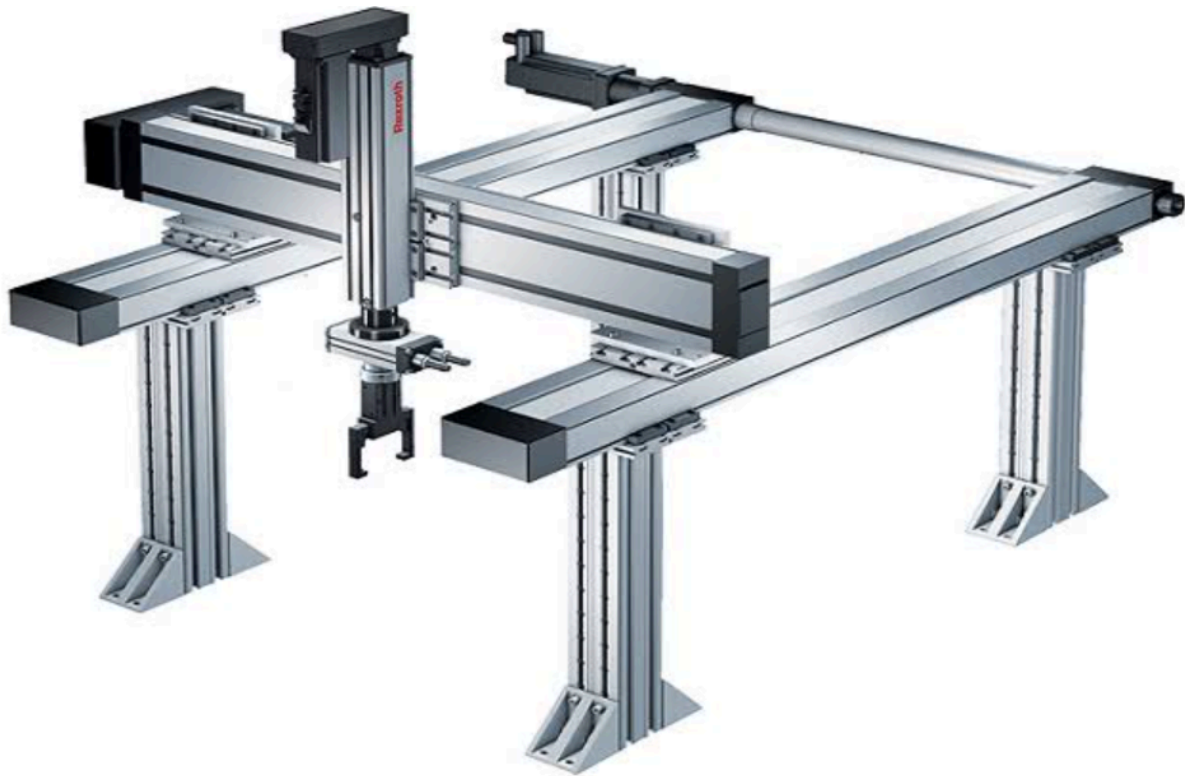


<https://scirobotics.com/petrisel/>

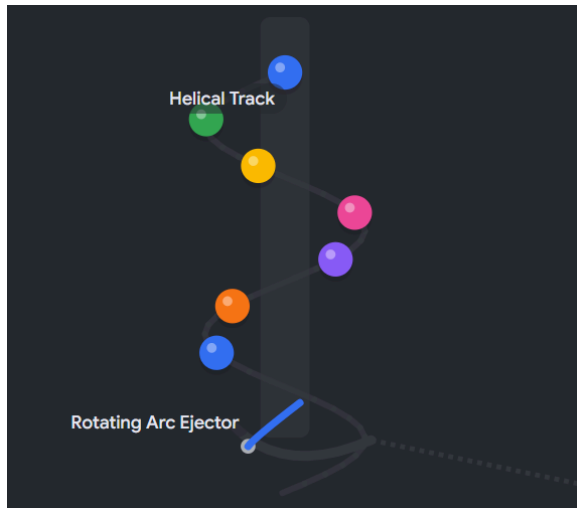


This is called the Cartesian Gantry Robot.

1. If we are using a phone instead of a camera, then we have to look into that. First of all, how much MP? Then what phone? What is the budget of the phone? How will we transfer the images from the phone? And if we transfer, will the image quality decrease? How to make the phone work in sync with our main processing system? Is the phone our main processing system? If yes, then how to do it???
2. Now comes the arm. Are we using an arm similar to scara with that horizontal axis motion that he was talking about? Or are we using a simple scara arm? Or are we planning on making a 3d printer similar looking arm(called cartesian gantry robot)? Then what gripper to use? Will it be similar to that 3 claw test tube holder? Can we convert that to 4 claws and then use it maybe? How is the petri dish gonna be held?

From sides or from top and bottom? How will we make the grippers soft so that there is no deformation of the petri dish? In the video that we saw, was there a lid on the petri dish? In some petri dishes, the lid doesn't cover the whole petri dish. So in the side surface, a tiny bit of petri dish is also exposed. So maybe, somehow, we use that kinda petri dish and we make such a gripper which is soft and can hold both petri dish and lid together from the sides. We will have to see this.

3. Need to see into that helical holder that utkarsh told for storing petri dishes.



First of all, in this, won't we need a really wide slide to hold these petri dishes? Because the diameter of this whole system is 2x the dia of petri dish + middle cylinder diameter. So this will consume a lot of space in storing the petri dishes because very few can actually fit as compared to when we are stacking them vertically. And there is also fear of some dish falling, this can create a lot of damage. And this is for when we are doing the analysis of petri dishes in any order. What we are thinking makes the system more advanced. In that, I can choose which petri dish to do the analysis for and then I won't need to go in some specific order. Do we need a specific order? How will the gripper hold it? Can't do it from the sides. How will we insert one claw under the petri dish when it is kept on a plain surface?

4. So we have got like 3 ideas. One is that carousel and just stacking vertically and then taking one out and then doing analysis and then keeping it in another stack. The issue is that the petri dish is fully covered with a lid, so this might now work in those cases. The best till now is the one in which petri dishes are being held by some holder, it's tilted so that no petri dish falls down and then we select and see which petri dish to see. But this also gives the advantage of easy gripping of the petri dish. The 3 claws one that we saw. The third idea is the one that utkarsh told. Need to see which is the best.
5. Answer to the question that subrat asked: Is the surrounding air contaminated when the lid of the petri dish is lifted? No, removing the lid does not contaminate the surrounding air. Instead, opening the dish exposes the sterile agar inside to whatever microbes are already floating in the room air.
6. Is the carousel gonna be controlled by hand or motor? I think it should just be motor. Or else if someone rotates the carousel, then we won't know what position we are in. Like i don't know whether we will be able to properly sense how much we have moved. So i think it should be motorised. Subrat talked about moving this by using gears and a stepper motor. Does gear increase torque? I don't think so. Also like

when the carousel is deaccelerating, there will be chances of petri dishes falling out. I think we can fix that by keeping the petri dishes at an angle. How much angle so that the sample doesn't get mixed up?

7. How many compartments in the carousel? Are we planning on storing all 1200 petri dishes in our compartment at once? Too much I guess.
8. Right now we are not going to think about the temp control, chimney, enclosure or all such stuff. These are just add ons that can be seen later.
9. Now about the orientation of the petri dish. Are we planning on qr or barcode? Well whatever we choose, how many of them should be there on the boundary of the petri dish? So the standard size of the petri dish is 90 mm x 15 mm, this is with the lid on. I think going with the bar code is a better option as the qr will be very small in this height. We won't even get full 15 mm, it will be around 12 mm or something.
10. For the outer and inner ring forming after some time, around the disc, the reason is:

why two rings form

- **The Inner Ring (Complete Inhibition):** This represents the area where the antibiotic concentration is high enough to completely kill or stop all bacterial growth.  www.zubairkhalid.com
- **The Outer Ring (Partial Inhibition):** This hazy, less-defined ring is caused by a sub-population of the bacteria that is slightly more tolerant, or by bacteria that managed to multiply a few times before the diffusing antibiotic reached a concentrated enough level to freeze them in place.  ResearchGate

So we need to measure the inner one, and not the outer one for our calculations.

11. Do we measure the maximum radius or the average one? So when the ring forming is cut because of the boundary, that's an issue caused by the person placing the antibiotics disks. These are called truncated. To measure the correct radius, we try to find some arc of the circle that isn't being interrupted by the outer edge of the petri dish and then measure from there. If that's not possible, then measure the maximum radius from the centre where the disc has been placed and double that for the approximate diameter.
12. https://commerce.bio-rad.com/webroot/web/pdf/inserts/CDG/en/Literature/inserts/660_98_01_2011_GB.pdf refer to this pdf for the names of the antibiotics. Like the code words written on the disk.
13. "For the disk diffusion test (also known as the Kirby-Bauer test), professional labs primarily use disposable plastic (polystyrene) petri dishes standardized to a 150 mm diameter." So according to what's written on google, we will be using plastic petri dish. But if we use plastic ones, there are chances of dishes getting deformed as the plastic bends.
14. TOA sensor. This is to know that we are approaching the petri dish and to slow down the speed of motion of the arm. The arm shouldn't collide with the carousel or the petri dish. The system should remain safe. SO we might use this sensor. But another thing, if by kinematics we know where we are in the global coordinates and we know the correct coordinates of the petri dish, the collision won't happen. This sensor seems like a precaution.

15. Why do I need qr? Like i come close to a sample and i need to know the sample id?
Or because I need to find a sample and i am seeing everywhere where that particular sample is?
A new petri dish came and I don't know where it is. So a camera to identify any changes.
We can use IR sensor
We can use push button, on each
16. X y motion - belt with stepper
Z motion - lead screw
Rotation - servo
Claw - servo
17. Garmina - Ai summary, Image processing, Research Paper, revised discussion that we had with prof subrat
18. Utkarsh - ubuntu in phone, hardware,